P-INDEPENDENT SETS, P-BASES AND THE DIFFERENTIALS OF A FIELD EXTENSION OF CHARACTERISTIC P

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Allowed standard prerequisites (editorial).

- p-independence and p-basis of a field extension, as defined immediately before the statement. definition environment at source lines 11338-11358 of the same chapter. Supplies the two properties that are being shown equivalent and the monomials x^E with exponents below p.
- Kaehler differentials of a field extension and the fact that dx_i form a K-basis when K is generated by the x_i over kK^p. used at proof lines 11394-11398 to derive (b) from (a). Provides the module in which the differentials live and the presentation K = kK^p[X_i]/(X_i^p - x_i^p) that makes the computation possible.
- Existence of k-derivations dual to a p-independent set, and the fact that every k-derivation kills kK^p. used at proof lines 11381-11387, announced inside the proof as theta_i with theta_i(x_j) = delta_ij. Produces the new relations that lead to the degree contradiction, which is the whole content of the first implication.

Source reference. Original lemma 07P2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9782–9782. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let K/k be a field extension. Assume k has characteristic $p > 0$. Let $\{x_i\}$ be a subset of K. The following are equivalent*

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(1) *the elements* $\{x_i\}$ *are* p*-independent over* k*, and*
(2) *the elements* dx_i *are* K*-linearly independent in* $\Omega_{K/k}$*.*

Any p*-independent collection can be extended to a* p*-basis of* K *over* k*. In particular, the field* K *has a* p*-basis over* k*. Moreover, the following are equivalent:*

(a) $\{x_i\}$ *is a* p*-basis of* K *over* k*, and*
(b) dx_i *is a basis of the* K*-vector space* $\Omega_{K/k}$*.*

Proof. Assume (2) and suppose that $\sum a_E x^E = 0$ is a linear relation with $a_E \in kK^p$. Let $\theta_i : K \to K$ be a k-derivation such that $\theta_i(x_j) = \delta_{ij}$ (Kronecker delta). Note that any k-derivation of K annihilates kK^p. Applying θ_i to the given relation we obtain new relations

$$\sum\nolimits_{E, e_i > 0} e_i a_E x_1^{e_1} \ldots x_i^{e_i - 1} \ldots x_n^{e_n} = 0$$

Hence if we pick $\sum a_E x^E$ as the relation with minimal total degree $|E| = \sum e_i$ for some $a_E \neq 0$, then we get a contradiction. Hence (1) holds.

If $\{x_i\}$ is a p-basis for K over k, then $K \cong kK^p[X_i]/(X_i^p - x_i^p)$. Hence we see that $\mathrm{d}x_i$ forms a basis for $\Omega_{K/k}$ over K. Thus (a) implies (b).

Let $\{x_i\}$ be a p-independent subset of K over k. An application of Zorn's lemma shows that we can enlarge this to a maximal p-independent subset of K over k. We claim that any maximal p-independent subset $\{x_i\}$ of K is a p-basis of K over k. The claim will imply that (1) implies (2) and establish the existence of p-bases. To prove the claim let L be the subfield of K generated by kK^p and the x_i. We have to show that $L = K$. If $x \in K$ but $x \notin L$, then $x^p \in L$ and $L(x) \cong L[z]/(z^p - x)$. Hence $\{x_i\} \cup \{x\}$ is p-independent over k, a contradiction.

Finally, we have to show that (b) implies (a). By the equivalence of (1) and (2) we see that $\{x_i\}$ is a maximal p-independent subset of K over k. Hence by the claim above it is a p-basis. □